

Mini Assessment

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1. If an API returns 429, what should your program do?

Answer: A 429 status code means “Too Many Requests” — the client has exceeded the API's rate limit. The program should check for a Retry-After header and wait before retrying. If there is no header, use exponential backoff, limit the number of retry attempts, log the event, and fail gracefully if retries are exhausted.

2. If `response.json()` fails, what should your program do?

Answer: `response.json()` can fail when the response body is empty, malformed, or not valid JSON. The program should catch the decoding error with `try/except`, check the response status and text, handle the error safely instead of crashing, and log the raw response for debugging.

3. If a JSON key is missing, why might `.get()` be safer than `[]`?

Answer: Using `data['key']` raises a `KeyError` when the key does not exist. Using `data.get('key')` is safer because it returns `None` (or a specified default) instead of raising an error. For example, `data.get('key', 'default_value')` provides a fallback and makes the program more robust.